Ramsey's theorem for pairs preserves a finite level scheme without controlling the jump

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Editorial note. The counted unit is the article's own Corollary cor:rt22-sig2 at source lines 1540-1547 together with the proof environment that begins on the next line (1548-1553); the pairing is the ordinary adjacent one.

Editorial note. The retained context is the statement of Proposition prop:rt22-sig2 at source lines 1506-1514, which the proof applies; that proposition is counted by a sibling card of this lane and the counted intervals are disjoint.

Editorial note. The counted proof carries the author citation of Belanger. The entry is supplied as a task-directory fragment holding it exactly as the bibtex run had formatted it, because the bibtex backend cannot be used here: the harness runs both xelatex passes before bibtex, so a generated bbl is never read back and the citation would stay undefined in the final pass.

Editorial note. The author has left a commented-out alternative first step at source lines 1550-1551; it is commented out and is not author text for our purposes, and it is not retained.

Editorial note. The retained preamble reproduces verbatim the author's document class at source line 2, his theorem declarations at source lines 16-25, his macro definitions at source lines 46-88 and 91, and his natural-operation macros at source lines 104-107, and it retains mathabx and cleveref; the stix-dependent LS2 block and the inputenc load with the utf8x option are not retained.

Editorial note. The retained preamble also loads the url package because the attached bibliography entry carries a url.

Allowed standard prerequisites (editorial).

- The finite-level proposition of this section. For a countable topped model of RCA_0 plus the scheme at level n minus one, every colouring and every auxiliary set above the jump of Y, there is a homogeneous set preserving the scheme at level n and controlling the jump; retained with this card at source lines 1506-1514.. The counted corollary is exactly this proposition with the auxiliary set supplied internally, so the statement must be available to see what is dropped.

- Omega-extension for finite levels of induction and collection. Every countable topped model of RCA_0-star plus the scheme at level n minus one can be omega-extended into a model of the same theory plus weak Konig's lemma, a result attributed to Harrington, Simpson-Smith and Hajek and recorded by Belanger; the citation is carried in the proof and its entry is attached with this card.. It replaces the external auxiliary set of the proposition by a set P obtained inside the model, which is the only new ingredient of the corollary.

- RCA_0-star and the jump of the topped model. RCA_0-star is the fragment of RCA_0 in which the extension step above is stated, and the model extended by its own jump satisfies that fragment at level n minus one because the scheme is assumed for the original model.. The proof applies the extension result to the model extended by its jump, which is why the corollary needs no external auxiliary set.

Proposition 0.1. *Fix $n \geqslant 2$ and let T_n be either $\mathsf{I}\Sigma^0_n$, or $\mathsf{B}\Sigma^0_{n+1}$. Consider a countable model $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + T_{n-1}$ topped by a set $Y \in S$. Then, for every coloring $f : [M]^2 \to 2$ in S and every set $P \gg Y'$ such that $\mathcal{M}[P] \models \mathsf{RCA}_0$, there exists $G \subseteq M$ such that*

1. *G is an M-infinite f-homogeneous set ;*
2. *$P \geqslant_T (G \oplus Y)'$;*
3. *$\mathcal{M}[G] \models \mathsf{RCA}_0 + T_n$.*

Corollary 0.2. *Fix $n \geqslant 2$ and let T_n be either $\mathsf{I}\Sigma^0_n$, or $\mathsf{B}\Sigma^0_{n+1}$. Consider a countable model $\mathcal{M} = (M, S) \models \mathsf{RCA}_0 + T_n$ topped by a set $Y \in S$. Then, for every coloring $f : [M]^2 \to 2$ in S, there exists $G \subseteq M$ such that*

1. *G is an M-infinite f-homogeneous set ;*
2. *$\mathcal{M}[G] \models \mathsf{RCA}_0 + T_n$.*

Proof. By Harrington; Simpson–Smith; Hajek (see Belanger [1]), every countable topped model of $\mathsf{RCA}^*_0 + T_{n-1}$ can be ω-extended into a model of $\mathsf{RCA}^*_0 + T_{n-1} + \mathsf{WKL}$. Since $\mathcal{M}[Y'] \models \mathsf{RCA}^*_0 + T_{n-1}$, there is some $P \subseteq M$ such that $P \gg Y'$ and $\mathcal{M}[P] \models \mathsf{RCA}^*_0 + T_{n-1}$. The conclusion follows from Proposition 0.1. □

References

[1] David R. Bélanger. Conservation theorems for the cohesiveness principle. 2022. Available at `https://arxiv.org/abs/2212.13011`.